V01.00

Mirror Volume I Overview

Observerhood from Constraint, Viability and Recursive Reliability

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Volume thesis

Observerhood is not primitive. It is a derived organisational condition in which a bounded physical or computational system preserves viability by maintaining world-models, self-models, reliability estimates, memory governance and physically supported continuity under perturbation.

Abstract

Volume I develops observerhood as the first substantive research arc of the Mirror Programme. It asks what must be true before a bounded system can be treated as an observer-like locus of modelled reality rather than a mere process, predictor, memory store or controller.

The volume answers that question in four layers. The Theory arc derives observerhood from bounded viability-constrained organisation. The Mathematics arc formalises constraint regimes, reliability salience, minimal observer thresholds and governed continuity. The Observerhood Labs test the computational role of reliability, repair and memory governance under perturbation. The Physics support arc asks what physical conditions are required for observer-like continuity to be implemented and preserved.

This overview is a map of Volume I only. It does not restate the full programme architecture or attempt to prove each technical claim. Its purpose is to state the volume's target, internal structure, boundaries and pressure points with minimal repetition.

Contents

1 Purpose and scope

This document is the volume-level overview for **Mirror Programme – Volume I: Observerhood**. MP-00 states the programme-level architecture. This document does something narrower: it explains the first research arc inside that architecture.

Volume I asks how observerhood can be derived from constraint, viability, recursive reliability and continuity. It does not treat the observer as a primitive subject, soul, witness or unexplained point of view. It also does not collapse observerhood into consciousness. The target is more precise: the organisational condition under which a bounded system maintains modelled reality from within a viability-constrained position.

Document discipline

MP-00 is the programme map. V01.00 is the map of Volume I. The underlying papers carry the arguments, proofs, experiments and physical-support analysis. This overview should therefore stay architectural, not encyclopaedic.

2 The observerhood question

The guiding question of Volume I is:

Guiding question

Under what conditions does a bounded physical or computational system become an observer-like locus of modelled reality?

The question is deliberately prior to standard questions about consciousness, perception or subjective experience. A system may process information without being an observer. It may predict its environment without locating itself inside that environment. It may store memory without governing which memories should become part of a continuing self-model. It may even describe itself without using self-model reliability to preserve viability.

Volume I therefore distinguishes five levels that are often conflated:

Level	Why it is insufficient by itself
Process	A process transforms states, but need not model a world or itself.
Predictor	A predictor forecasts states, but need not preserve a self-located organisation.
Self-model	A self-model represents the system, but may be unreliable, passive or inert.
Memory store	A memory store preserves records, but may also preserve false or identity-poisoning records.
Observer-like system	An observer-like system uses world-models, self-models, reliability, repair and memory governance to preserve continuity under constraint.

The volume's central contribution is to make the last category explicit.

3 Volume I in one page

The compressed spine of Volume I is as follows.

3.1 Core formal object

The basic formal object is a constraint regime:

$$C = (S, T, I, V, \kappa),$$

where S is a state space, T is an admissible transition structure, I is a family of invariants, V is a viability or persistence functional, and κ is a bounded capacity parameter.

This object does not make a system an observer. It supplies the grammar within which observerhood can be evaluated.

3.2 Core reliability threshold

Reliability becomes observer-relevant only when it improves expected viability after cost:

$$\Sigma_L(R_L, \rho) = \mathbb{E}[V_L \mid M_E, M_L, R_L, \rho] - \mathbb{E}[V_L \mid M_E, M_L, \pi_0] - K(R_L, \rho).$$

The threshold condition is:

$$\Sigma_L(R_L, \rho) > 0.$$

Here M_E is an environmental model, M_L is a local self-model, R_L is a reliability variable over the self-model, ρ is a repair or recalibration policy, π_0 is a baseline policy and K is the cost of maintaining and acting on reliability information.

3.3 Core continuity claim

Observerhood is not exhausted by a momentary threshold. Persistent observerhood requires governed continuity:

$$C_L^{\mathrm{cont}}(t, t+n) > \theta_C.$$

Continuity is not sameness and not raw memory storage. It is the governed preservation of self-model, reliability history, commitments and action-relevant memory under perturbation.

3.4 Physical support claim

Observerhood must be implemented somewhere. The physics support arc therefore treats an observer as a local constraint-maintenance process embedded in physical conditions that permit causal order, finite propagation, record formation, energetic maintenance, repair and eventual dissolution.

4 Architecture of Volume I

Volume I is organised into four arcs. Each arc has a distinct role, and the overview should not blur them.

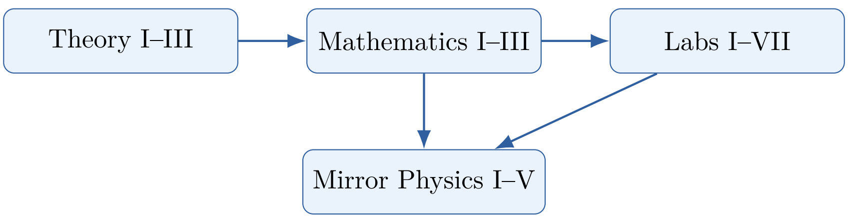

Figure 1: Volume I architecture. Theory defines the target; mathematics formalises the conditions; laboratories test the mechanisms; physics analyses support conditions.

Arc	Function
Theory	Derives observerhood as a recursive viability-constrained organisation and clarifies its philosophical consequences.
Mathematics	Turns the theory into constraint regimes, salience thresholds, observer predicates and continuity functionals.
Labs	Tests whether reliability, repair and memory governance improve viability in controlled artificial systems.
Physics support	Asks what physical support conditions are required for observer-like continuity to be implemented and maintained.

5 Paper map

This section states the internal structure of Volume I. It is a conceptual map of how the volume is organised.

5.1 Theory arc

Code	Paper	Role
V01.01	*Mirror Theory I: A Minimal Computational Theory of Recursive Observerhood*	Derives recursive observerhood from distinguishability, dynamics, viability and bounded computational capacity.
V01.02	*Mirror Theory II: Observerhood, Identity and Reality*	Develops the philosophical consequences of derived observerhood for identity, persistence, meaning, objectivity and realism.
V01.03	*Mirror Theory III: Recursive Observerhood in Context*	Situates Mirror Theory among predictive processing, active inference, autopoiesis, enactivism, self-model theory, global workspace theory, IIT and artificial intelligence.

5.2 Mathematics arc

Code	Paper	Role
V01.04	*Mirror Mathematics I: A Constraint Calculus for Recursive Observerhood*	Defines constraint regimes, viability functionals, bounded modelling, self-models, reliability variables and salience gain.
V01.05	*Mirror Mathematics II: Minimal Observer Thresholds*	Formalises the point at which self-model reliability becomes represented, action-guiding, cost-sensitive and positively coupled to viability.
V01.06	*Mirror Mathematics III: Identity, Continuity and Memory Governance*	Extends the threshold account from momentary observerhood to persistent observerhood through governed self-memory incorporation.

5.3 Computational observerhood labs

Lab	Title	Function
Lab I	*Recursive Self-Model Reliability*	Tests whether self-model reliability improves viability under self-relevant perturbation.
Lab II	*Channel-Decomposed Reliability*	Shows that decomposing reliability channels is useful for attribution but insufficient without effective repair.
Lab III	*Actionable Reliability and Cost-Sensitive Repair*	Tests whether reliability becomes useful only when it changes action and repair is worth its cost.
Lab IV	*Minimal Reliability Thresholds*	Sweeps perturbation rate, repair cost and reliability threshold to identify positive and negative viability regions.
Lab V	*Recursive Reliability*	Tests when reliability over reliability becomes useful under estimator corruption.
Lab VI	*Neuro-Symbolic Persistence*	Extends the architecture to language-like agents with symbolic commit gates and channel reliability.
Lab VII	*Autobiographical Continuity*	Tests whether commit-gated memory governance protects identity continuity under false memory, goal drift and identity injection.

5.4 Physics support arc

Code	Paper	Role
V01.07	*Mirror Physics I: Constraint Propagation and the Physical Conditions for Observerhood*	Examines causal and physical support for local constraint-maintenance processes.
V01.08	*Mirror Physics II: Constraint Salience and Physical Coupling*	Asks when a physical constraint becomes relevant to an observer-like system.
V01.09	*Mirror Physics III: Thermodynamic Observerhood*	Examines the energetic cost of modelling, memory, repair and persistence.
V01.10	*Mirror Physics IV: Causal Invariance and the Speed of Constraint Propagation*	Studies causal limits on modelling, coordination and repair.
V01.11	*Mirror Physics V: Entropy, Time and Constraint Dissolution*	Analyses decay, temporal ordering and the dissolution of maintained constraint organisation.

6 Claims and boundaries

6.1 Claims

Volume I claims that:

1. observerhood can be studied without treating observers as primitive;
2. self-modelling is insufficient unless self-model reliability becomes action-guiding and viability-relevant;
3. reliability tracking has costs and thresholds, and is not a universal performance booster;
4. persistent observerhood requires memory governance rather than mere storage;
5. computational demonstrations can test the viability role of reliability, repair and continuity mechanisms without claiming to demonstrate consciousness; and
6. physical implementation matters because observer-like continuity must be causally, energetically and materially supported.

6.2 Non-claims

Volume I does not claim that:

1. consciousness has been fully explained;
2. subjective experience is reducible to a single variable;
3. all predictive, self-referential or memory-bearing systems are observers;
4. the labs prove artificial general intelligence or moral status;
5. the mathematics replaces existing mathematical frameworks; or
6. the physics support arc overturns established physics.

6.3 Boundary discipline

The boundary is simple. A system is not observer-like merely because it contains information about itself. Under the Mirror predicate, self-information becomes observer-relevant only when it is embedded in a bounded organisation where reliability, repair and memory governance help preserve viability and continuity under perturbation.

7 What the laboratories show

The Observerhood Labs are important because they prevent the theory from remaining purely verbal. Their pattern is deliberately mixed.

Lab I shows that self-model reliability can produce a strong advantage when the perturbation attacks the agent's represented relation to itself. Lab II shows that decomposed reliability is not enough by itself. Lab III adds actionability and cost-sensitive repair. Lab IV identifies threshold regions where repair is useful and regions where it is not. Lab V shows that recursive reliability is useful only when first-order reliability can itself fail in viability-relevant ways. Lab VI extends the framework to language-like symbolic commitments. Lab VII shows that memory becomes continuity only when candidate memories are filtered before becoming durable self-memory.

Laboratory lesson

The labs do not show that reliability always helps. They show a narrower and stronger point: reliability matters when it is channel-appropriate, action-guiding, cost-sensitive and coupled to viability or continuity under the right perturbation.

8 Relation to existing frameworks

Volume I is adjacent to predictive processing, active inference, cybernetics, autopoiesis, enactivism, self-model theory, global workspace theory, integrated information theory, intentional systems theory and contemporary artificial intelligence.

Its contribution is not to replace those frameworks. Its contribution is to isolate a prior structural target: the conditions under which a bounded system becomes an observer-like locus of modelled reality at all.

In this sense, Volume I sits between several extremes. Against passive realism, it treats world-appearance as modelled by bounded systems. Against unconstrained idealism, it insists that modelling is constrained by structures not invented by the observer. Against behaviour-only accounts, it requires explicit self-model reliability and repair. Against consciousness-first accounts, it treats phenomenal consciousness as a later and harder question rather than an initial primitive.

9 Pressure points

Volume I should be judged by whether it makes observerhood more precise. The main pressure points are:

1. whether observerhood can be distinguished from ordinary prediction, control and memory;
2. whether reliability salience can be operationalised rather than merely named;
3. whether minimal thresholds identify real boundaries rather than labels;
4. whether computational results remain robust under different environments and costs;
5. whether memory governance provides a useful account of continuity without over-rigidity;
6. whether the physics support arc remains respectful of established physical constraints; and
7. whether the framework avoids becoming a universal metaphor.

The volume is strengthened if the same vocabulary identifies real distinctions, failures, thresholds and repair mechanisms. It is weakened if every system becomes observer-like merely because the vocabulary can be applied to it after the fact.

10 Conclusion

Volume I develops observerhood as a derived condition of bounded systems that preserve viability by maintaining models, reliability estimates, repair policies, governed memory and physically supported continuity.

The Theory arc defines the conceptual target. The Mathematics arc turns that target into formal conditions. The Observerhood Labs test the computational role of reliability, repair and memory governance. The Physics support arc asks what implementation conditions make observer-like continuity possible.

The volume's discipline is narrow but important: observerhood must do work. It must identify why a process is not yet an observer, why a predictor is not enough, why memory is not continuity, and why reliability matters only when it changes action in a viability-relevant way.

If Volume I succeeds, it does not solve every problem of consciousness, identity or physics. It establishes a controlled foundation for asking those questions without assuming the observer at the beginning.